

# Jayden Dongwoo Lee

Postdoctoral Researcher, AITA InnoCORE, Department of Aerospace Engineering, KAIST

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## Research Interests

Guidance, navigation, and control (GNC) for aerial vehicles that remain safe under model uncertainty, component faults, and loss of GNSS, validated on flight hardware.

- **AI-based guidance and control:** Gaussian processes, SINDy, and reinforcement learning for adaptive, incremental, and model-predictive control.
- **Fault diagnosis and fault-tolerant control:** data-driven FDD with SINDy and Koopman models, control allocation for multirotors, eVTOL, and UAM.
- **GNSS-denied navigation:** factor graph optimization with INS, UWB, terrain, and quantum magnetometer measurements.
- **Future directions:** digital twins and integrated validation; certifiable AI flight control (handling-qualities requirements, run-time assurance).

## Education

### Ph.D. in Aerospace Engineering

Sep. 2021 – Aug. 2025

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

- Dissertation: *Data-Driven Fault Diagnosis and Fault-Tolerant Control for Redundancy of eVTOL Systems*
- Outstanding Ph.D. Dissertation Award, KAIST College of Engineering (Feb. 2026)
- Advisor: Prof. Hyochoong Bang GPA 3.88/4.3

### M.S. in Aerospace Engineering

Sep. 2019 – Aug. 2021

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

- Thesis: *Model Reference Adaptive Control Design for Quadrotor UAV Transporting Unknown Payloads*
- Advisor: Prof. Hyochoong Bang GPA 3.91/4.3

### B.S. in Mechanical and Aerospace Engineering, Cum Laude

Mar. 2015 – Aug. 2019

Ulsan National Institute of Science and Technology (UNIST)

Ulsan, Korea

- Thesis: *Design of Hybrid Propulsion Multirotor System* Research advisor: Prof. Hungsun Son GPA 3.62/4.3

## Professional Experience

### Postdoctoral Researcher

Sep. 2025 – Present

AITA InnoCORE Program, Department of Aerospace Engineering, KAIST

Daejeon, Korea

- AI-based fault-tolerant control and quantum-magnetometer-aided navigation for eVTOL and unmanned aerial systems (advisor: Prof. Hyochoong Bang).
- Mentoring graduate and undergraduate students; flight-test campaigns with INS, RTK-GPS, and magnetometer payloads.

### AI Research Engineer (concurrent with graduate studies)

Apr. 2021 – Feb. 2024

AI Research Center, Hanyoung Nux Co., Ltd.

Korea

- Industrial temperature-controller algorithms combining PID tuning with data-driven adaptation; productized control modules.

## Honors and Awards

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- 2026 Outstanding Ph.D. Dissertation Award, KAIST College of Engineering
- 2025 Best Paper Award, Korea Institute of Military Science and Technology (KIMST)
- 2025 Best Paper Award, Society for Aerospace System Engineering (SASE)
- 2025 Paan Foundation Academic Scholarship; Cho Jung-hoon Scholarship, KAIST; Moral Character Scholarship, KAIST
- 2024 Top Excellence Award, Defense-Technology-Based Startup Competition, Defense Acquisition Program Administration (DAPA)
- 2024 Excellence Prize (team), Challenge! K-Startup Grand Finale, Ministry of SMEs and Startups
- 2024 Grand Paper Award and Best Paper Award, Institute of Control, Robotics and Systems (ICROS)
- 2024 Best Paper Award ( $\times 2$ ), Society for Aerospace System Engineering (SASE)
- 2024 Best Paper Award, Korean Society for Aeronautical and Space Sciences (KSAS)
- 2024 Excellent Teaching Assistant Award, KAIST
- 2023 Best Paper Award, 23rd International Conference on Control, Automation and Systems (ICCAS)
- 2020 Best Paper Award, Society for Aerospace System Engineering (SASE)
- 2019 Cum Laude, UNIST

## Publications

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### International Journal Papers (24)

- [1] **Jayden Dongwoo Lee**, Hyochong Bang, "Data-Driven Fault Detection and Diagnosis for eVTOL Systems Using SINDy," *IEEE Transactions on Aerospace and Electronic Systems*, 2026. (EARLY ACCESS) [doi]
- [2] Byeonghyeon Jo, **Jayden Dongwoo Lee**, Hyochong Bang, "Reinforcement Learning-Based Error-Gated  $\sigma$ -Modification Adaptive Control for Cascade Missile Longitudinal Autopilots under Aerodynamic Uncertainty," *International Journal of Control, Automation, and Systems*, 2026. (ACCEPTED)
- [3] Bohyun Chang, **Jayden Dongwoo Lee**, Hyochong Bang, Donghoon Lee, "Robust Point Cloud Registration using LiDAR Intensity Levels for On-orbit Proximity Operations," *Journal of Aerospace Information Systems*, 2026. (ACCEPTED)
- [4] Minwoo Kang, **Jayden Dongwoo Lee**, Seonghun Yun, Hyochong Bang, "Sliding Mode Disturbance Observer-based Robust Model Predictive Control using Sparse Identification of Nonlinear Dynamics," *Journal of Aerospace Information Systems*, 2026.
- [5] Hyeonmin Kim, **Jayden Dongwoo Lee**, Bohyun Chang, Hyochong Bang, "Adaptive Sliding Window Factor Graph Optimization with Multi-tag UWB/INS Fusion for Multirotor UAVs," *Aerospace Science and Technology*, 2026.
- [6] **Jayden Dongwoo Lee**, Yoonseong Kim, Youngjae Kim, Hosun Lee, Ji-Hun Cha, Hyochong Bang, "Sensor and Actuator Fault Detection and Isolation for Urban Air Mobility," *IEEE Sensors Journal*, 2026.
- [7] Lamsu Kim, Hosun Lee, **Jayden Dongwoo Lee**, Hyochong Bang, "Gaussian Process-Driven Incremental Flight Dynamics Control and Center of Gravity Adjustment for Tailsitter UAVs," *Journal of Intelligent & Robotic Systems*, 2026. [doi]
- [8] Yujin Lee, Juhyeong Park, **Jayden Dongwoo Lee**, Hyochong Bang, "Attitude Takeover Control of Combined Spacecraft with Actuator Saturation using Prescribed Performance and Time-Delay Estimation," *International Journal of Aeronautical and Space Sciences*, 2026.
- [9] Dongshik Won, Kyungwoo Hong, **Jayden Dongwoo Lee**, Heoncheol Lee, Hyochong Bang, "In-Orbit Validation of a Heterogeneous COTS Architecture for Onboard AI: Characterizing System-Level Vulnerabilities and Operational Failure Pathways," *International Journal of Aeronautical and Space Sciences*, 2026.
- [10] Gilsu Park, Chun-Gon Kim, Heechul Kwon, **Jayden Dongwoo Lee**, Jinwoo Park, Yoonkook Son, Jong Guk Kim, Ji-Hun Cha, "Tailored Stitching and Vertical Stacking for High-Voltage Multifunctional Structural Batteries with Enhanced Electrochemical-Mechanical Coupling," *Advanced Science*, 2025.
- [11] Dohyun Kim, Seungho Lee, **Jayden Dongwoo Lee**, Hohyeong Lee, Hyochong Bang, "Guided Exploration Reinforcement Learning for 3D UAV Pursuit-Evasion Games," *International Journal of Control, Automation, and Systems*, 2025.

- [12] **Jayden Dongwoo Lee**, Youngjae Kim, Yoonseong Kim, Hyochoong Bang, "Sparse Identification of Nonlinear Dynamics-based Model Predictive Control for Multirotor Collision Avoidance," *IET Control Theory & Applications*, 2025.
- [13] Juhyeong Park, **Jayden Dongwoo Lee**, Hyochoong Bang, "Vibration Suppression Adaptive Prescribed Performance Control for Flexible Spacecraft with Uncertain Moment of Inertia," *Acta Astronautica*, 2025.
- [14] **Jayden Dongwoo Lee**, Donghoon Seo, Jongho Shin, Ji-Hun Cha, Hyochoong Bang, "Fast Real-Time State-of-Health Estimation Method for Lithium-ion Battery using Sparse Identification of Nonlinear Dynamics," *Journal of Intelligent & Robotic Systems*, 2025.
- [15] Ji-Hun Cha, **Jayden Dongwoo Lee**, Tae-Hyun Kim, Jong Guk Kim, Yoonkook Son, Chun-Gon Kim, "Structural Battery Composites with Interlayer-Bonding Stitching Architecture for Maintaining Minimal Electrode Gap," *Composites Part B: Engineering*, 2025.
- [16] **Jayden Dongwoo Lee**, Youngjae Kim, Lamsu Kim, Natnael S. Zewge, Hyochoong Bang, "Sparse Online Gaussian Process Regression-based Robust Nonlinear Dynamics Inversion for Multirotor with Forward Flight Ground Effect," *Aerospace Science and Technology*, 2025.
- [17] Dain Yoon, Kyung-Mi Na, **Jayden Dongwoo Lee**, Chang-Hun Lee, Hyochoong Bang, "Practical Fault-Tolerant Control Allocation based on Attainable Control Set Analysis for a Coaxial Dodecaopter," *Control Engineering Practice*, 2025.
- [18] **Jayden Dongwoo Lee**, Lamsu Kim, Kwangwoo Jang, Seongheon Lee, Hyochoong Bang, "Feedback Linearization-based Model Reference Adaptive Control for Multirotor UAVs under Uncertainties from Fuel and Payload," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, 2025.
- [19] Deok-Young Lee, Gabin Yun, Minseok Choi, KyuHyeon Cho, **Jayden Dongwoo Lee**, Dong-Hoon Lee, "Miniaturized Dual-Photodiode Sensor for Simultaneous Wavelength and Irradiance Measurement across the 500–1000 nm Range," *Applied Optics*, 2024.
- [20] Lamsu Kim, **Jayden Dongwoo Lee**, Seongheon Lee, Hyochoong Bang, "Fault Detection for Re-initialization of Online Gaussian Process Regression Using Kernel Linear Independence Test," *International Journal of Control, Automation, and Systems*, 2024.
- [21] **Jayden Dongwoo Lee**, Lamsu Kim, Seongheon Lee, Hyochoong Bang, "Data-driven Fault Diagnosis of Nonlinear Systems with Parameter Uncertainty using Deep Koopman Operator and Weighted Window Extended Dynamic Mode Decomposition," *International Journal of Control, Automation, and Systems*, 2024.
- [22] **Jayden Dongwoo Lee**, Sukjae Im, Lamsu Kim, Hyunjoo Ahn, Hyochoong Bang, "Data-driven Fault Detection and Isolation for Multirotor System using Koopman Operator," *Journal of Intelligent & Robotic Systems*, 2024.
- [23] Lamsu Kim, **Jayden Dongwoo Lee**, Natnael S. Zewge, Hyochoong Bang, "Gaussian Process-based State Derivative Estimator with Temporal Input in Incremental Flight Control Design," *Aerospace Science and Technology*, 2024.
- [24] **Jayden Dongwoo Lee**, Lamsu Kim, Natnael S. Zewge, Hyochoong Bang, "Fault-tolerant Control for Aircraft with Structural Damage using Sparse Online Gaussian Process Regression," *International Journal of Aeronautical and Space Sciences*, 2024.

#### Domestic Journal Papers (6)

- [1] **Jayden Dongwoo Lee**, Gunmo Lee, Minseok Choi, Hyochoong Bang, Deok-Young Lee, "MagNav-based Navigation Simulation Using Magnetic Anomaly Maps for GPS-denied Environments," *Journal of Institute of Control, Robotics and Systems*, vol. 32, no. 4, pp. 460–467, 2026. [doi]
- [2] **Jayden Dongwoo Lee**, Kyungwoo Hong, Chulsoo Lim, Hyochoong Bang, Dongju Lim, Daesung Park, Kihoon Song, "A Design of Model Predictive Control and Nonlinear Disturbance Observer-based Backstepping Sliding Mode Control for Terrain Following," *Journal of the Korea Institute of Military Science and Technology*, vol. 27, no. 4, pp. 495–506, 2024.
- [3] Hohyung Lee, Lamsu Kim, **Jayden Dongwoo Lee**, Kwangwoo Jang, Kyoungyong Jo, Taesan Park, Hyochoong Bang, "Development of Look Angle Guidance Law for Loitering Munition under Velocity Change," *Journal of the Korean Society for Aeronautical & Space Sciences*, vol. 51, no. 9, pp. 617–625, 2023.

- [4] Lamsu Kim, **Jayden Dongwoo Lee**, Hohyung Lee, Suwoon Hong, Hyochoong Bang, "Modeling of Fixed Wing UAV and Flight Control Computer Based Autopilot System Development for Integrated Simulation HILS Environment," *Journal of the Korean Society for Aeronautical & Space Sciences*, vol. 50, no. 12, pp. 857–866, 2022.
- [5] **Jayden Dongwoo Lee**, Lamsu Kim, Kwangwoo Jang, Seongheon Lee, Hyochoong Bang, "Model Reference Adaptive Control of a Quadrotor Considering the Uncertainty of Payload," *Journal of the Korean Society for Aeronautical & Space Sciences*, vol. 49, no. 9, pp. 749–757, 2021.
- [6] Lamsu Kim, **Jayden Dongwoo Lee**, Hyochoong Bang, "Maneuvering Target Tracking with 3D Variable Turn Model and Kinematic Constraint," *Journal of the Korean Society for Aeronautical & Space Sciences*, vol. 48, no. 11, pp. 881–888, 2020.

#### International Conference Papers (30)

- [1] **Jayden Dongwoo Lee**, Minseok Choi, Deok-Young Lee, Daham Kim, Sin Hyuk Yim, Sangkyung Lee, Yoontae Hwang, Hyochoong Bang, "Federated Extended Kalman Filter Integrating Terrain-Referenced and Quantum Magnetometer-Aided Navigation for GNSS-Denied Environments," *ION 2026 Pacific PNT Meeting, Honolulu, USA*, 2026.
- [2] Yujin Lee, Juhyeong Park, **Jayden Dongwoo Lee**, Hyochoong Bang, "Attitude takeover control of postcapture combined spacecraft with adaptive prescribed performance," *11th EUROPEAN CONFERENCE FOR AERONAUTICS AND SPACE SCIENCES (EUCASS), Rome, Italy*, 2025.
- [3] Hyeonmin Kim, **Jayden Dongwoo Lee**, Hyochoong Bang, "Adaptive Sliding Window Factor Graph Optimization For Enhanced UWB/INS Integrated UAV Localization," *11th EUROPEAN CONFERENCE FOR AERONAUTICS AND SPACE SCIENCES (EUCASS), Rome, Italy*, 2025.
- [4] Jiho Han, **Jayden Dongwoo Lee**, Hyochoong Bang, "Dual Virtual Target-Based Polynomial Curve Fitting Guidance for Urban Air Mobility Path Following," *AIAA SCITECH*, 2025.
- [5] Bohyun Chang, **Jayden Dongwoo Lee**, Hyochoong Bang, "Key Feature Detection for Rendezvous and Proximity Operation Using BRDF Segmentation Algorithm," *AIAA SCITECH*, 2025.
- [6] Juhyeong Park, **Jayden Dongwoo Lee**, Hyochoong Bang, "Adaptive Prescribed Performance Attitude Control for Flexible Spacecraft using Observer," *75th International Astronautical Congress (IAC), Milan, Italy*, 2024.
- [7] **Jayden Dongwoo Lee**, Youngjae Kim, Yoonseong Kim, Seungho Lee, Sanghyeok Park, Seongmin Kim, Shin Yuseop, Hyochoong Bang, "Sparse Identification of Nonlinear Dynamics-based Model Predictive Control for Multirotor Collision Avoidance," *2024 International Conference on Robot Intelligence Technology and Applications (RiTA)*, 2024.
- [8] **Jayden Dongwoo Lee**, Hyeonmin Kim, Yoonseong Kim, Hyochoong Bang, "Koopman Fault-Tolerant Model Predictive Control for Multirotor with Actuator Fault," *The 24th International Conference on Control, Automation and Systems (ICCAS), Yeosu, Korea*, 2024.
- [9] Minwoo Kang, **Jayden Dongwoo Lee**, Seonghun Yun, Hyochoong Bang, "Data-Driven Model Discovery and Control of High-Performance Trainer Aircraft via Sparse Identification of Nonlinear Dynamics," *34th Congress of the International Council of the Aeronautical Sciences (ICAS), Florence, Italy*, 2024.
- [10] Hosun Lee, **Jayden Dongwoo Lee**, Hyochoong Bang, "Aerodynamic Model Identification of a VTOL Tailsitter UAV Using Sparse Identification of Nonlinear Dynamics," *34th Congress of the International Council of the Aeronautical Sciences (ICAS), Florence, Italy*, 2024.
- [11] **Jayden Dongwoo Lee**, Sukjae Im, Hyochoong Bang, "Data-driven fault detection and isolation for quadrotor using sparse identification of nonlinear dynamics and Thau observer," *2024 International Conference on Unmanned Aircraft Systems (ICUAS)*, 2024.
- [12] Dohyun Kim, Seungho Lee, **Jayden Dongwoo Lee**, Hohyeong Lee, Hyochoong Bang, "Guided Exploration Reinforcement Learning for UAV Pursuit-Evasion Games," *The 23rd International Conference on Control, Automation and Systems (ICCAS), Yeosu, Korea*, 2023.
- [13] **Jayden Dongwoo Lee**, Lamsu Kim, Hosun Lee, Hyochoong Bang, "Data-Driven Fault Detection and Identification of Nonlinear Systems Using Weighted Window Extended Dynamic Mode Decomposition," *The 23rd International Conference on Control, Automation and Systems (ICCAS), Yeosu, Korea*, 2023.

- [14] Lamsu Kim, **Jayden Dongwoo Lee**, Hyochoong Bang, "Fault Diagnosis using Kernel Linear Independence Test in Reproducing Kernel Hilbert Space," *The 23rd International Conference on Control, Automation and Systems (ICCAS)*, Yeosu, Korea, 2023.
- [15] **Jayden Dongwoo Lee**, Hyungjoo Ahn, Jaeho Lee, Hyochoong Bang, "Reinforcement learning-based nonlinear disturbance observer for UAV with parametric uncertainty and unmodeled dynamics," *AIAA SCITECH*, 2023.
- [16] Jaeho Lee, Junwoo Park, **Jayden Dongwoo Lee**, Hyochoong Bang, "Sequential Cooperative Reinforcement Learning of Mothership Routing Problem," *AIAA SCITECH*, 2023.
- [17] **Jayden Dongwoo Lee**, Minwoo Kang, Seungjun Han, Hyochoong Bang, "Terrain Following of Tiltrotor UAV using MPC and Nonlinear Disturbance Observer-Based Backstepping," *22nd International Conference on Control, Automation and Systems (ICCAS)*, Busan, Korea, 2022.
- [18] **Jayden Dongwoo Lee**, Kwangwoo Jang, Seongheon Lee, Lamsu Kim, Hyochoong Bang, "Closed-Loop MRAC Augmented LQR with Integral Action for Quadrotor UAV under the Uncertainties," *6th CEAS Conference on Guidance, Navigation and Control (EuroGNC)*, Berlin, Germany, 2022.
- [19] **Jayden Dongwoo Lee**, Lamsu Kim, Chulsoo Lim, Hyochoong Bang, "Incremental Nonlinear Dynamic Control for UAM with Coaxial Rotors Performance Degradation," *2022 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Niigata, Japan, 2022.
- [20] Chulsoo Lim, Kwangwoo Jang, **Jayden Dongwoo Lee**, Hyochoong Bang, "Toward reliable UAM autonomous flight: Path planning under dynamic obstacle detection using vision," , 2022.
- [21] Kwangwoo Jang, **Jayden Dongwoo Lee**, Hyochoong Bang, "Model Reference Adaptive Controller for the Quad Tiltrotor Unmanned Aerial Vehicle in Hovering and Transition mode using Radial Basis Function Neural Network," , 2022.
- [22] **Jayden Dongwoo Lee**, Lamsu Kim, Hyochoong Bang, "Fault-Tolerant Control for Aircraft with Structural Damage using Gaussian Process Regression," *33rd Congress of the International Council of Aeronautical Sciences (ICAS)*, Stockholm, Sweden, 2022.
- [23] Hyungjoo Ahn, Chulsoo Lim, **Jayden Dongwoo Lee**, Hyochoong Bang, "Safe Reinforcement Learning Based Multi-Rotor Collision Avoidance with Unexpected Obstacles," *33rd Congress of the International Council of Aeronautical Sciences (ICAS)*, Stockholm, Sweden, 2022.
- [24] Lamsu Kim, **Jayden Dongwoo Lee**, Hyochoong Bang, "Incremental Nonlinear Dynamic Control with Sparse Non-parametric Bayesian Regression of Damaged Aircraft," *33rd Congress of the International Council of Aeronautical Sciences (ICAS)*, Stockholm, Sweden, 2022.
- [25] **Jayden Dongwoo Lee**, Youngjae Kim, Sukjae Kim, Hyochoong Bang, "Reliability study of redundant flight control system for urban air mobility using voting algorithm," *9th International Conference on Engineering Failure Analysis (ICEFA)*, Shanghai, China, 2022.
- [26] **Jayden Dongwoo Lee**, Lamsu Kim, Jaeho Lee, Hyochoong Bang, "Design for a Dual-Redundant Flight Control Computer Using a Model Predictive Voting Algorithm," *9th International Conference on Robot Intelligence Technology and Applications (RiTA)*, Daejeon, Korea, 2021.
- [27] **Jayden Dongwoo Lee**, Lamsu Kim, Jaeho Lee, Hyochoong Bang, "Gaussian process-based feedback linearization control for quad-tiltrotor," *2021 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Jeju, Korea, 2021.
- [28] Hyungjoo Ahn, Kwangwoo Jang, **Jayden Dongwoo Lee**, Junwoo Park, Kyungwoo Hong, Hyochoong Bang, "Experimental Model of Coaxial Rotors Performance Degradation Induced by Wake Effect," *2021 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Jeju, Korea, 2021.
- [29] Lamsu Kim, **Jayden Dongwoo Lee**, Hanseok Park, Hyochoong Bang, "Nonlinear Dynamic Inversion Control for Aircraft with Disturbance Observer based on Gaussian Process Regression," *2021 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Jeju, Korea, 2021.
- [30] **Jayden Dongwoo Lee**, Chulsoo Lim, Seongheon Lee, Hyochoong Bang, Hyukjoon Kwon, Seungwook Lee, Woulsun Choi, Yeongil Choi, "Dynamic Performance Analysis of Hybrid-Electric Propulsion Systems for an Urban Air Mobility," *9th EUROPEAN CONFERENCE FOR AERONAUTICS AND SPACE SCIENCES (EUCASS)*, Lille, France, 2021.

## Domestic Conference Papers (57)

- [1] **Jayden Dongwoo Lee**, 김동현, 조병현, et al., "Online SINDy 기반 비선형 모델 예측 제어," 제어로봇시스템학회 국내학술대회, 2026.
- [2] **Jayden Dongwoo Lee**, Natnael S. Zewge, 이덕영, et al., "한국형 양자 자기장 센서를 활용한 위성 궤도 결정 개념 연구," 한국항공우주학회 학술발표회, 2026.
- [3] 조병현, 김동현, 이호선, et al., "모델 불확실성이 존재하는 미사일의 제어 장벽 함수 기반 다중 모델 참조 적응 제어," 한국항공우주학회 학술발표회, 2026.
- [4] **Jayden Dongwoo Lee**, Deok-Young Lee, Hyochoong Bang, "Factor Graph Optimization-Based Magnetic Navigation Using a Quantum Magnetometer in GNSS-Denied Environments," *Proc. IPNT Conference (Institute of Positioning, Navigation, and Timing)*, 2026.
- [5] **Jayden Dongwoo Lee**, 이정우, 강민우, et al., "항공·우주 분야에서의 양자 센서의 활용 방안 및 기술 개발 사례 연구," 한국항공우주학회 학술발표회, 2025.
- [6] 김동현, 이호선, **Jayden Dongwoo Lee**, et al., "가용 조종력 집합을 고려한 틸트로터 항공기의 구동기 고장 상황에서의 비행 제어 가능성 분석," 한국항공우주학회 학술발표회, 2025.
- [7] **Jayden Dongwoo Lee**, 김동현, 이호선, et al., "eVTOL 시스템을 위한 데이터 기반 고장 진단 기법 설계," 한국항공우주학회 학술발표회, 2025.
- [8] 조병현, 이유진, **Jayden Dongwoo Lee**, et al., "모델 불확실성이 존재하는 환경에서의 미사일의 성능 보장 제어," 항공우주시스템공학회 학술대회, 2025.
- [9] 고만준, Natnael S. Zewge, **Jayden Dongwoo Lee**, et al., "심우주 탐사를 위한 최적 삼각측량 기반 광학 항법 연구," 한국항공우주학회 학술발표회, 2025.
- [10] 이호선, **Jayden Dongwoo Lee**, 김동현, et al., "이단계 칼만 필터 기반 분산형 전기 추진 비행체의 구동기 고장 진단 및 고장 허용 제어," 항공우주시스템공학회 학술대회, 2025.
- [11] 이유진, 박주형, **Jayden Dongwoo Lee**, et al., "시간 지연 추정 기법을 이용한 결합 우주비행체의 성능 보장 자세 제어," 한국항공우주학회 학술발표회, 2025.
- [12] 강민우, **Jayden Dongwoo Lee**, 윤성훈, et al., "물리정보 SINDy를 통한 고정익 항공기의 공력 모델링 및 슬라이딩 모드 외란 관측기를 통한 강인한 비선형 모델예측제어 기법 설계," 한국항공우주학회 학술발표회, 2025.
- [13] 김현민, **Jayden Dongwoo Lee**, 장보현, et al., "거리 기반 적응형 슬라이딩 윈도우 팩터그래프 UAV 위치추정 기법," 한국항공우주학회 학술발표회, 2025.
- [14] **Jayden Dongwoo Lee**, 방효충, "SINDy를 이용한 데이터 기반의 센서 및 구동기 고장 진단 및 분리," 한국항공우주학회 학술발표회, 2024.
- [15] 한지호, **Jayden Dongwoo Lee**, 방효충, "바람 환경에서의 도심항공교통(UAM) 경로 추종 유도," 한국항공우주학회 학술발표회, 2024.
- [16] 김현민, **Jayden Dongwoo Lee**, 방효충, "미식별 선박 탐지를 위한 국내·외 RF 감시·정찰 위성 기술 개발 사례 연구," 한국항공우주학회 학술발표회, 2024.
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## Patents (Applications)

- [1] Koopman-operator-based linear observer for multirotor motor fault diagnosis, KR 10-2024-0143755 (application published 2025).
- [2] Deep-Koopman-operator parity-space fault diagnosis for nonlinear systems, KR 10-2024-0146601 (application published 2025).
- [3] Flight-vehicle fault diagnosis via kernel linear independence test in RKHS, KR 10-2024-0171982 (application published 2025).
- [4] Triple-redundant flight control computer with an integrated voter, KR 10-2025-0026443 (application published 2025).

## Teaching and Mentoring

### Teaching Assistant

2021 – 2025

Department of Aerospace Engineering, KAIST (Excellent TA Award, 2024)

Daejeon, Korea

- AE30000 Flight Dynamics Project; AE350 Aerospace Control Engineering; AE552 Advanced Linear Stability and Control; AE40005 Spacecraft Systems

### Graduate Mentor, Hanwha-KAIST Space Pebble Program

2022 – 2025

KAIST Department of Aerospace Engineering / Hanwha Aerospace

Daejeon, Korea

- Mentored middle-school teams through aerospace mission projects across all four cohorts.

### Research Mentor, MiliTech Challenge

2021 – 2024

Center for Research Officers for National Defense, KAIST

Daejeon, Korea

- Mentored team won the Grand Prize (Minister of National Defense Award) at the 5th MiliTech Challenge (2024); another team received an Encouragement Prize at the 2024 Korea Air Force Academy Future Aerospace Conference.

## Technical Skills

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|                      |                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------|
| Programming          | C/C++, Python, MATLAB/Simulink                                                                         |
| ML / control         | PyTorch, TensorFlow, scikit-learn, CVXPY, CasADi, GPyTorch, PySINDy                                    |
| Flight / embedded    | PX4, ArduPilot, LattePanda, NVIDIA Jetson, STM32, Red Pitaya (FPGA-DSP), FCC design (OrCAD), HILS/PILS |
| Navigation / sensors | VectorNav VN-200, u-blox ZED-F9P RTK, optically pumped magnetometer                                    |
| Design / simulation  | CATIA, SolidWorks, Gazebo, AirSim, X-Plane                                                             |

## References

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